Class 1: Preparation for Learning Edge Networks

XG, VS Code, GitHub, JSON, AI, and 5G Toolbox

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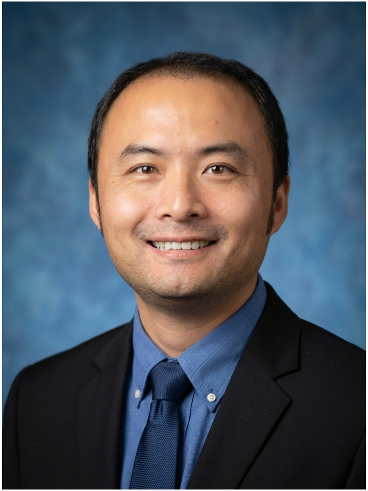
CIS 4930/5930 Future Edge Networks
https://xinliulab.github.io/FSU-CIS4930-CIS5930-Future-Edge-Networks/



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Research:
AI-driven edge networks and intelligent wireless systems

Hobby: *Not Working*
Started at FSU in Fall 2024
Click here to visit my homepage

Mobile has made a leap every 10 years

1G → **2G** → **3G** → **4G** → **5G**

1G	2G	3G	4G	5G
Voice	Text	Web	Video	Connected things

1980s 1990s 2000s 2010s 2020s

Each generation changed what people expected a phone to do. The device, radio link, and battery budget moved together.

A ringtone as a memory cue

The Nokia tune marks the 2G moment.

- Ringtones made the phone personal.
- Polyphonic audio showed that phones had become small media devices.
- A familiar sound carried a decade of mobile culture.

Instructor note: every time I hear this tune, I remember my youth, the beautiful days, and the people who made those days bright.

Play the Nokia ringtone memory.

Applications follow capability

The phone got stronger.

- More radio throughput
- More local processing
- Better screens, sensors, and storage
- More energy demand

The default app changed.

- 1G: voice on analog cellular links
- 2G: text, contact lists, and ringtones
- 3G: web, email, and early video calls
- 4G: maps, streaming, and short video
- 5G: connected things, sensing, and edge AI

A new “G” matters when phones, radios, and nearby computing support a new daily habit.

The phone leaves the wall.

- Analog cellular voice
- Expensive devices and sparse coverage
- The core experience: call from a car or street

Memory cue: the Motorola DynaTAC "brick phone."
Image: Redrum0486, Wikimedia Commons, CC BY-SA.

2G: Text goes mobile

The phone becomes personal.

- Digital voice and SMS
- SIM cards, contacts, and prepaid plans
- Ringtones, Snake, and the always-on pocket device

Memory cue: Nokia-style durable feature phones.

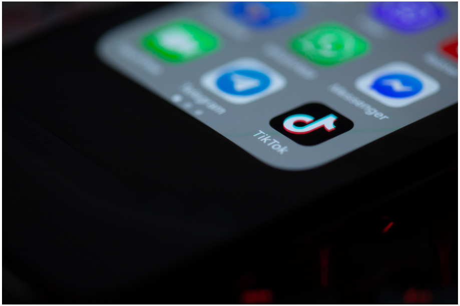
Image: Jpk, Wikimedia Commons, CC BY-SA.

Data becomes the product.

- Mobile web, email, and app stores
- Camera phones and multimedia messages
- Early video calls over UMTS networks

Memory cue: video call as a new phone behavior.

4G: Video becomes normal



The phone becomes a video network.

- LTE makes mobile broadband feel routine
- Streaming, ride sharing, maps, and live apps
- Short video turns uplink and downlink into daily traffic

Memory cue: TikTok and short-video apps.
Image: Solen Feyissa, Wikimedia Commons, CC BY-SA.

Connectivity moves into systems.

- Enhanced mobile broadband
- Low-latency control and edge computing
- Sensing, vehicles, factories, and AI services

Memory cue: active antenna units and dense radio sites.

The next “G” should make us ask a different question.

- If every device can run or reach a large language model, what should the network provide?
- Which tasks should stay on the device, move to nearby edge compute, or go to the cloud?
- What new applications appear when wireless links carry model state, sensor context, and real-time feedback?

Think of 6G as a network for intelligence, sensing, and control.

A 6G-style product question: WiCi One

Local AI needs nearby compute.

- A phone has limited heat, battery, and GPU budget.
- Cloud AI adds latency, cost, privacy risk, and connectivity dependence.
- Nearby wireless GPU compute gives another design point.

WiCi One example

- GPU over Wi-Fi 7
- Real-time private AI
- Phones and laptops stay light
- Local endpoint for AI apps

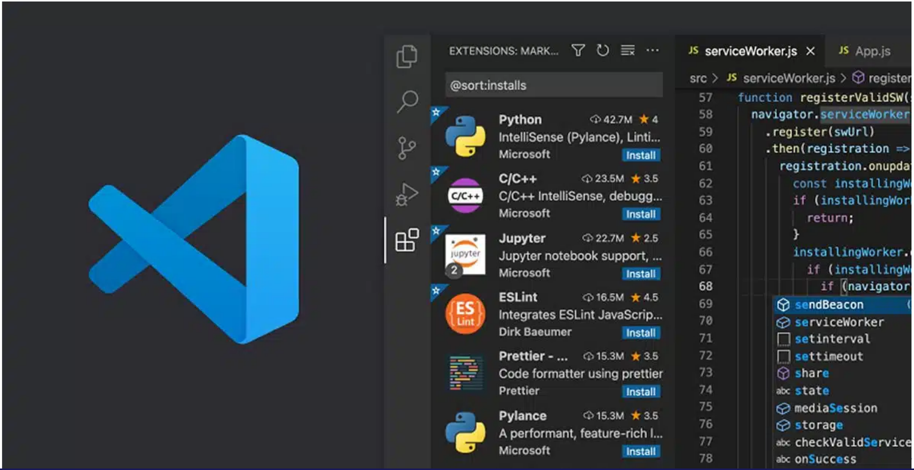
wici.ai/wici-one

What industry appears when wireless serves AI?

- Personal AI that follows the user across devices
- Wireless GPU sharing for phones, laptops, glasses, and robots
- Sensing systems that understand places in real time
- Local-first AI services for homes, labs, factories, and vehicles

Your question for this course: what should the network become when models become part of daily life?

The Best IDE in the World:
Visual Studio Code



Just remember one thing:

`Ctrl + Shift + P`

This shortcut opens the Command Palette — your gateway to: *Extensions, settings, themes, terminal, and everything else.*

“Professor... I don’t want to install all that stuff.”

GitHub Resources

No problem!
We use **GitHub Codespaces!**
No installs. No setup. Just code in your browser.

We will use GitHub in three key ways:

- **GitHub Codespaces** — Write and run code directly in your browser. No local setup needed.
- **GitHub Classroom** — Submit your programming assignments.
- **Code Version Control** — Learn how to use GitHub to manage your code, especially when collaborating in a team.

We Use JSON for Assignment Answers

- JSON (JavaScript Object Notation) is a lightweight, structured data format based on **key–value pairs**.
- It's simple, human-readable, and widely used in modern software systems (e.g., APIs, configs, logs).
- In the AI and cloud era, JSON has become a **de facto standard**.
- For assignments, JSON lets us automatically parse your answers and process them programmatically.

What does JSON look like?

- JSON is a structured data format based on **key–value pairs**.
- JSON supports **nesting** — values can be strings, numbers, lists, or even other JSON objects.

Example JSON Format

```
{
  "student_id": "FSU12345",
  "answers": {
    "Q1": "B",
    "Q2": "D"
  }
}
```

My View on AI

Can we use **ChatGPT, Copilot, Gemini, etc.**
to help with assignments?

Absolutely —
I strongly
encourage it!

Use It!

- You’re welcome to use AI tools to brainstorm, generate code, or write explanations.
- You can even include notes like “This answer was inspired by GitHub Copilot” — that’s fine.
- I want you to learn how to use AI productively, just like in real-world engineering.

But Outsmart It.

- My job is to make sure assignments are designed so that AI won't give you perfect answers easily.
- Your real skill is not in typing a prompt — it's in understanding, adapting, and improving what AI gives you.

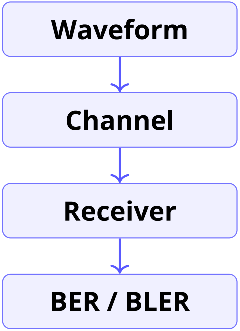
AI is welcome. Lazy copying is not.

MATLAB 5G Toolbox

Do we still need to build 5G from scratch?

Start from a reference.

MATLAB already gives you
a solid 5G NR toolbox.



What Is 5G Toolbox?

- A MATLAB add-on to **model, simulate, analyze, and test** 5G communications systems.
- Built-in workflows for:
    - **Waveform generation**
    - **Link-level simulation**
    - **Golden reference design verification**
    - **System-level simulation**
- More details here

Waveforms in Minutes

- Generate uplink and downlink waveforms with:
  - **subcarrier spacing**, **bandwidth parts**, **frame numerology**
- Wireless Waveform Generator app:
  - **NR Test Models (NR-TM)**
  - **Downlink Fixed Reference Channels (FRC)**
  - For **FR1** and **FR2**

Channels and Signals You Will See in Real 5G

- Physical channels:
  - Uplink/downlink shared channels
  - Control and broadcast channels
- Synchronization and reference signals:
  - PSS, SSS
  - DM-RS (demodulation, timing, sync)
  - Phase tracking, CSI-related reference signals
  - SRS (sounding reference signal)
  - PRACH (random access)

Link-Level Simulation: The Classic Pipeline

- End-to-end link simulation:
  - transmitter $\rightarrow$ channel $\rightarrow$ receiver
- Standard channel models:
  - **CDL** and **TDL** channel models
- Metrics you can compute:
  - BER, BLER, throughput
- Receiver building blocks:
  - channel/timing estimation, synchronization, equalization

Coding and Procedures Are Included

- Channel coding components:
  - **LDPC** and **Polar** coding subcomponents
- Initial access procedures:
  - cell search and selection
  - decode **MIB** and **SIB1**
  - construct SS bursts and sync using DM-RS

System-Level Simulation: Multi-Node, Scheduling

- Multi-node simulations at the system level
- Example: **PUSCH scheduling** under different MAC strategies
- Great for understanding:
  - RAN behaviors, resource allocation, and performance tradeoffs

The Part I Really Want You to Use

It is open MATLAB source code.

Read it like a textbook.

Why This Matters

- You can access features as **customizable MATLAB source code**.
- Treat it as a **golden reference** for design verification.
- If you dissect the implementation step by step (with the help of your GPT teacher),
  - you will understand how 5G NR really works.
  - **you will become a 5G (6G) engineer faster**.

Takeaways

- We are living in the **AI era**.
- With AI, the distance between you and “top talent” is much smaller than you think.
- Knowledge is no longer the bottleneck. **Curiosity is.**
- Your advantage is the ability to **ask good questions** and **analyze what you get**.